



Jyothy Charitable Trust®

Jyothy Institute of Technology

Tataguni, off Kanakapura Road, Bengaluru – 560082

Approved by The All India Council for Technical Education (AICTE) - New Delhi.

Affiliated to Visvesvaraya Technological University (VTU), Belagavi.

ECE, CSE, ISE, CIV - Accredited by National Board of Accreditation (NBA) - New Delhi.

INTRODUCTION TO WEB DEVELOPMENT

Dates and Branches Covered:

- **28th, 29th, and 30th November 2024:** Computer Science and Civil Engineering students.
- **2nd and 3rd December 2024:** Artificial Intelligence and Machine Learning (AIML) students.
- **3rd and 4th January 2025:** Information Science and Engineering students.

Workshop Objective: The workshop aimed to provide first-year students with foundational knowledge and skills in web development. By the end of the sessions, participants were expected to understand essential concepts in networking, web design, and programming, equipping them with practical tools to develop and deploy basic web applications. This initiative also sought to inspire interest in web technologies as a potential career or academic path. This workshop also had an objective that each student needs to build their own portfolio web pages.

1. Basics of Networking

- Introduction to computer networks: key components and functions.
- Detailed explanation of the 3-way handshake process in TCP connections with real-world analogies to simplify understanding.
- Importance of networking in web development.

2. HTML

- Understanding the structure of a web page.
- Introduction to essential HTML tags, attributes, and semantic elements.
- Building a basic web page with headings, paragraphs, images, and links.

3. CSS

- The role of CSS in enhancing web page aesthetics.
- Overview of CSS selectors, properties, and values.
- Practical exercises on inline, internal, and external stylesheets.
- Creating visually appealing layouts using CSS techniques.

4. CDN (Bootstrap)

- Introduction to Content Delivery Networks and their advantages.
- Hands-on demonstration of Bootstrap for responsive web design.
- Practical examples using Bootstrap components like navigation bars, grids, and buttons.
- Customizing Bootstrap themes for unique designs.

5. JavaScript

- Basics of JavaScript syntax and its role in web interactivity.
- Integrating JavaScript with HTML and CSS.
- Practical exercises on event handling, conditional statements, and basic DOM manipulation.
- Building simple interactive features like form validation and dynamic content updates.

Workshop Structure:

- Each session began with a brief introduction to the day's topics, followed by detailed explanations and practical demonstrations.
- Participants engaged in hands-on coding exercises to reinforce theoretical concepts.
- Mini-projects were introduced progressively, with each topic contributing to a final integrated project.
- Dedicated Q&A sessions addressed doubts and provided additional insights into best practices.

Key Highlights:

- Active participation from students across all branches.
- Visual aids and real-world examples made complex concepts, such as the TCP 3-way handshake, accessible and engaging.
- The use of Bootstrap components significantly boosted confidence in creating responsive designs quickly.
- Interactive exercises in JavaScript captivated participants, showcasing the power of dynamic web development.

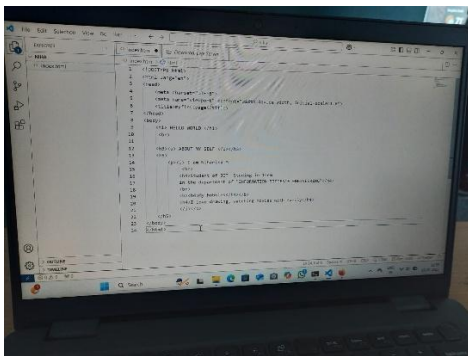
Feedback and Outcomes:

- **Positive Feedback:**
 - Students appreciated the structured approach and the balance between theory and practice.
 - Many expressed confidence in creating basic, functional web pages by the end of the workshop.
 - The inclusion of Bootstrap and its practical usage received specific praise for its immediate applicability.
- **Suggestions for Improvement:**
 - Some students requested deeper dives into JavaScript, including advanced topics such as AJAX, APIs, and backend integration.
 - Participants recommended extending the workshop duration to allow for more comprehensive project-building sessions.

Conclusion: The workshop successfully introduced first-year students to the fundamentals of web development, creating a strong foundation for future learning and exploration. The enthusiasm and feedback from participants highlight the effectiveness of the workshop and the demand for similar initiatives. The hands-on approach ensured that students not only learned concepts but also gained confidence in applying them practically.

Acknowledgments: We extend our gratitude to the **HoD's Dr. Madhusudhan sir and Dr. Vishwanth sir, faculty coordinators, technical support team, and student volunteers** for their invaluable contributions to the success of this workshop.

SNAPSHOTS





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WORKSHOP ON JAVASCRIPT AND WEB TECHNOLOGIES

Date: 04th and 05th July 2025

Resource Persons: B S Shreesha, Anirudh G Bharadwaj, Sriraksha S

Target Branches:

1. Artificial Intelligence and Machine Learning (AIML)
2. Computer Science and Engineering (CSE)
3. Civil Engineering (CV)
4. Information Science and Engineering (ISE)

OBJECTIVES

The objective of this workshop was to introduce the foundational concepts of web development with a strong focus on JavaScript and the supporting technologies that drive interactive websites. The aim was to equip students with hands-on experience and confidence in using HTML, CSS, Bootstrap, and JavaScript to build dynamic web applications.

TOPICS COVERED

1. HTML (HyperText Markup Language)

- Structure of a webpage
- Tags and elements
- Forms, Inputs, Tables, Lists, and Semantic elements
- Best practices for writing clean HTML

2. CSS (Cascading Style Sheets)

- Styling fundamentals (selectors, properties, values)
- Box model and layout techniques
- Class vs ID selectors
- CSS positioning and display properties
- Responsive design using media queries

3. CDN (Content Delivery Network) - Bootstrap

- Introduction to Bootstrap and its CDN usage

- Grid system and layout design
- Components: Navbar, Cards, Buttons, Modal, Alerts
- Bootstrap utilities for spacing, alignment, colors
- Responsiveness and mobile-first design

4. JavaScript (JS)

- Introduction to JavaScript and its role in the web
- Syntax and data types
- Variables (let, const, var)
- Operators and control structures (if, switch, loops)

5. DOM (Document Object Model)

- Understanding the DOM tree
- Accessing elements via getElementById, querySelector, etc.
- Modifying content and styles dynamically
- Creating and removing DOM elements
- Use of innerHTML, textContent

6. Events in JavaScript

- Event types: click, submit, mouseover, keydown, etc.
- Event listeners and handlers
- Event bubbling and delegation
- Form validation and interactivity

7. Arrays

- Array declaration, initialization, and iteration
- Array methods: push(), pop(), shift(), unshift(), slice(), splice(), map(), filter(), forEach()
- Practical use cases in web apps (e.g., storing form inputs)

8. Objects

- Object syntax and properties
- Nesting and accessing properties
- Methods inside objects
- Working with arrays of objects
- Application: storing structured data (e.g., user profiles, products)

Methodology

- Hands-on sessions with live coding examples
- Problem-solving activities and mini-projects
- Use of real-world examples to explain concepts
- Interactive Q&A sessions to ensure conceptual clarity
- Encouraged students to build mini web applications during the workshop

Learning Outcomes

By the end of the workshop, students were able to:

- Design and structure webpages using HTML and CSS
- Use Bootstrap via CDN to rapidly develop responsive UI components
- Write JavaScript code to add interactivity and logic
- Manipulate the DOM to create dynamic interfaces
- Handle user input using event listeners
- Use arrays and objects to manage and display data dynamically
- Understand how front-end technologies work together in modern web development

Quiz Session

As part of learning reinforcement, a **quiz** was conducted at the end of the workshop to evaluate students' understanding of the topics discussed.

Quiz Details:

- **Format:** Online (Multiple Choice Questions)(Multiple Select Questions)
- **Topics Covered:** HTML, CSS
- **Participation:** Students from all four branches participated enthusiastically
- **Platform:** Google Forms

17. Which of the following are valid input types in HTML? *

- ☐ text
- ☐ Password
- ☐ checkbox
- ☐ select

18. Which attributes are commonly used with <input> elements? *

- ☐ type
- ☐ name
- ☐ value
- ☐ title

Sample Questions given in the quiz

7/5/2025 15:09:32	jayaprakash9th@gmail.c	28 / 30	1JT24AI014	JAYA PRAKASH B Y
7/5/2025 15:09:33	kantambatlushripad@gr	27 / 30	1JT24AI040	shripad kantambatlu
7/5/2025 15:09:47	shreyasbharadwaj78@g	23 / 30	1JT24AI039	Shreyas S Bharadwaj
7/5/2025 15:09:53	yashasm19@gmail.com	30 / 30	1JT24AI054	Yashas M
7/5/2025 15:09:55	nishchaysgowda@gmail	22 / 30	1JT24AI026	Nishchay s gowda
7/5/2025 15:09:57	muralidhartadela@gmai	29 / 30	1JT24AI048	Pavan
7/5/2025 15:10:07	amoghbhat.n1a2@gmai	29 / 30	1JT24AI003	Amogha N Bhat
7/5/2025 15:10:09	honnashreem0405@gm	30 / 30	1JT24AI013	Honnashree M
7/5/2025 15:10:10	shreeyashaarvari@gmai	30 / 30	1JT24AI035	Shaarvari. M
7/5/2025 15:10:12	salmas3953@gmail.con	30 / 30	1JT24AI034	SALMA S

Results – AIML

7/4/2025 14:18:43	bhanumathi3780@gmail	29 / 30	1JT24CS059	Hemashree.B	Computer Science and E B
7/4/2025 14:18:50	nirmalathimmaraju89@q	29 / 30	1JT24CS058	Harshitha T	Computer Science and E B
7/4/2025 14:18:55	gowthamirgowthami996	30 / 30	1jt24cs048	Gowthami	Computer Science and E B
7/4/2025 14:19:38	ranjanapal9035@gmail.c	24 / 30	1JT24CS018	Anjali pal	Computer Science and E A
7/4/2025 14:20:09	arjunsomesh9876@gma	26 / 30	1JT24CS022	Arjun Somesh M	Computer Science and E A
7/4/2025 14:20:18	2akhilagd@gmail.com	26 / 30	1JT24CS011	Akhila G.Dambal	Computer Science and E A
7/4/2025 14:20:21	farhanashafi817@gmail	27 / 30	1JT24CS045	Farhana Shafi	Computer Science and E A
7/4/2025 14:20:22	ayushbunty12@gmail.cc	25 / 30	1JT24CS024	Ayush G	Computer Science and E A
7/4/2025 14:20:23	abhishekjavali26@gmail	27 / 30	1JT24CS003	Abhishek Javali	Computer Science and E A
7/4/2025 14:20:23	h404127@gmail.com	27 / 30	1JT24CS053	Harish K	Computer Science and E B
7/4/2025 14:20:29	ayushisarkhel@gmail.co	26 / 30	1JT24CS026	Ayushi Sarkhel	Computer Science and E A
7/4/2025 14:20:30	abhayprakash9945@gm	27 / 30	1JT24CS0001	ABHAY PRAKASH	Computer Science and E A
7/4/2025 14:20:30	darshandachu512@gma	26 / 30	1JT24CS035	Darshan S	Computer Science and E A

Results – CSE A

7/4/2025 15:16:22	nandinihegde004@gmail	28 / 30	1JT24CS093	Nandini Hegde	Computer Science and E B
7/4/2025 15:16:23	manojkumrk1234@gma	24 / 30	1JT24CS080	Manoj kumar k	Computer Science and E B
7/4/2025 15:16:25	nath40764@gmail.com	26 / 30	1JT24CS079	Manjunath s	Computer Science and E B
7/4/2025 15:16:30	punithc5883@gmail.con	29 / 30	1JT24CS115	Punith c	Computer Science and E C
7/4/2025 15:16:49	mounikadrmounikadr96i	28 / 30	1JT24CS089	Mounika D R	Computer Science and E B
7/4/2025 15:17:19	monikakn511@gmail.co	20 / 30	1JT24CS086	Monika K N	Computer Science and E B
7/4/2025 15:18:59	kalpithamahadev76@gn	27 / 30	1JT24CS068	KALPITHA M	Computer Science and E B
7/4/2025 15:18:59	laasya2804@gmail.com	28 / 30	1JT24CS071 aas	LAASYA N SUVARNA	Computer Science and E B
7/4/2025 15:19:00	hitaishii2005@gmail.cor	27 / 30	1JT24CS060	HITAISHI G	Computer Science and E B
7/4/2025 15:19:05	nagashreer74@gmail.c	25 / 30	1JT24CS092	Nagashree RR	Computer Science and E B
7/4/2025 15:19:09	hiteshav2006@gmail.co	24 / 30	1JT24CS061	Hitesh Av	Computer Science and E B
7/4/2025 15:22:51	manasamanasa1063@q	25 / 30	1JT24CS78	Manasa K	Computer Science and E B

Results – CSE B

7/4/2025 12:56:23	tusharprao35@gmail.co	24 / 30	1JT24CS166	Tushar P Rao	Computer Science and E D
7/4/2025 12:56:43	chinnichinni16038@gma	25 / 30	1JT24CS146	Shwetha.R	Computer Science and E D
7/4/2025 12:56:50	sinchus240567@gmail.c	27 / 30	1JT24CS149	Sinchana s	Computer Science and E D
7/4/2025 12:56:50	gowdasrashmi7@gmail.	28 / 30	1JT24CS122	Rashmi S	Computer Science and E C
7/4/2025 12:57:12	someshm2006@gmail.c	28 / 30	1JT24CS152	SOMESH M	Computer Science and E D
7/4/2025 12:57:12	tarundevanga225@gma	28 / 30	1JT24CS160	Tarun M	Computer Science and E D
7/4/2025 12:57:29	sirishamc93@gmail.con	22 / 30	1JT24CS150	Sirisha	Computer Science and E D
7/4/2025 12:57:30	ruqaiyahzahara@gmail.c	27 / 30	1JT24CS125	Ruqaiyah Zahara	Computer Science and E C
7/4/2025 12:57:37	tharunvj12345@gmail.c	23 / 30	1JT24CV007	Tharun k	Civil Engineering D
7/4/2025 12:57:45	supradasahu@gmail.cor	26 / 30	1JT24CS156	Suprada Sahu	Computer Science and E D
7/4/2025 12:57:55	yashoda.patil15@gmail.	28 / 30	1JT24CS183	Yashodha S Patil	Computer Science and E D

Results – CSE C and D

7/5/2025 16:25:22	sam.shummi@gmail.cor	28 / 30	1JT24IS038	Samhita Bharadwaj I	Information Science and F
7/5/2025 16:25:22	keerthannaka@gmail.co	28 / 30	1JT24IS024	KEERTHANA K A	Information Science and F
7/5/2025 16:25:26	anaghaml1108@gmail.c	28 / 30	1JT24IS004	Anagha ML	Information Science and F
7/5/2025 16:25:26	rockabhay07@gmail.cor	21 / 30	1JT24IS001	ABHAY A SHARMA	Information Science and F
7/5/2025 16:25:45	harshini.mr.2006@gmail	29 / 30	1JT24IS026	M R HARSHINI	Information Science and F
7/5/2025 16:25:54	bhoomikakulkarni104@ç	26 / 30	1JT24IS009	Bhoomika Kulkarni	Information Science and F
7/5/2025 16:25:58	pranavbangala26@gmai	26 / 30	1JT24IS033	PRANAV V BANGALA	Information Science and F
7/5/2025 16:26:00	bsr7527@gmail.com	25 / 30	1JT24IS040	Shreyas R Bharadwaj	Information Science and F
7/5/2025 16:26:08	shakhapurechandana@ç	25 / 30	1JT24IS010	Chandana Chandrakant	Information Science and F
7/5/2025 16:26:12	yaminib074@gmail.com	26 / 30	1JT24IS053	Yamini B	Information Science and F
7/5/2025 16:26:12	rakshitha1832006@gme	18 / 30	1JT24IS037	Rakshitha s	Information Science and F
7/5/2025 16:26:12	gaganvn.321@gmail.cor	26 / 30	1JT24IS014	Gagan V N	Information Science and F
7/5/2025 16:26:14	ganaviyogesh486@gma	26 / 30	1JT24IS016	Ganavi Y	Information Science and F

Results – ISE

Conclusion

The workshop was highly effective in introducing essential front-end web development skills to first-year students. The practical approach helped demystify JavaScript and encouraged students from various branches to explore further learning in web technologies. This session laid a solid foundation for future courses and projects involving UI/UX and web development.

Session Photos



